Deliverable 5: Verification Requirements

System Verification Requirements

Verification Requirements.docx

VR.2.1 Dual System Capture

The system shall be tested to confirm synchronized capture of eye-tracking and scene video streams at a minimum sampling rate of **200 Hz** by measuring frame count over a 10-second interval.

Rationale:

Quality Score: 0%

Labels: *None to display.*

VR.2.2 Data Synchronization

The system shall be tested to confirm timestamp alignment between eye and scene data streams with a maximum deviation of **10 milliseconds** by comparing timestamps across recorded frames.

Rationale:

Quality Score: 0%

Labels: *None to display.*

VR.2.3 Operational Gaze Accuracy

The system shall be tested to confirm gaze estimation accuracy within **±2 degrees** during inspection using a 9-point calibration grid under typical indoor factory conditions.

Rationale:

Quality Score: 0%

Labels: *None to display.*

VR.2.4 Data Upload

The system shall be tested to confirm the successful upload of inspection session data to the cloud within **10 seconds** under standard network conditions.

Rationale:

Quality Score: 0%

Labels: *None to display.*

VR.2.5 Inspection Analytics

The system shall be demonstrated to generate gaze heatmaps and scan path visualizations aligned with predefined inspection zones using recorded inspection data.

Rationale:

Quality Score: 0%

Labels: *None to display.*

VR.2.6 Battery Performance

The system shall be tested to confirm continuous operation for a minimum of **6 hours** under normal inspection usage conditions.

Rationale:

Quality Score: 0%

Labels: *None to display.*

VR.2.7 Calibration Accuracy

The system shall be tested to confirm calibration accuracy within **±1 degree** using a controlled calibration procedure with predefined fixation targets.

Rationale:

Quality Score: 0%

Labels: *None to display.*

VR.2.8 Inspection Coverage

The system shall be tested to confirm the accurate calculation of inspection coverage percentage by comparing detected gaze zones with predefined inspection zones.

Rationale:

Quality Score: 0%

Labels: *None to display.*

VR.2.9 Hands-Free Operation

The system shall be tested to confirm the accurate calculation of inspection coverage percentage by comparing detected gaze zones with predefined inspection zones.

Rationale:

Quality Score: 0%

Labels: *None to display.*

VR.2.10 Inspection Report

The system shall be demonstrated to generate inspection reports containing heatmaps, scanpaths, and inspection coverage metrics from recorded inspection sessions.

Rationale:

Quality Score: 0%

Labels: *None to display.*

VR.2.11 Network Reliability

The system shall be tested to confirm successful data upload in at least **95% of attempts** under standard network conditions across multiple trials.

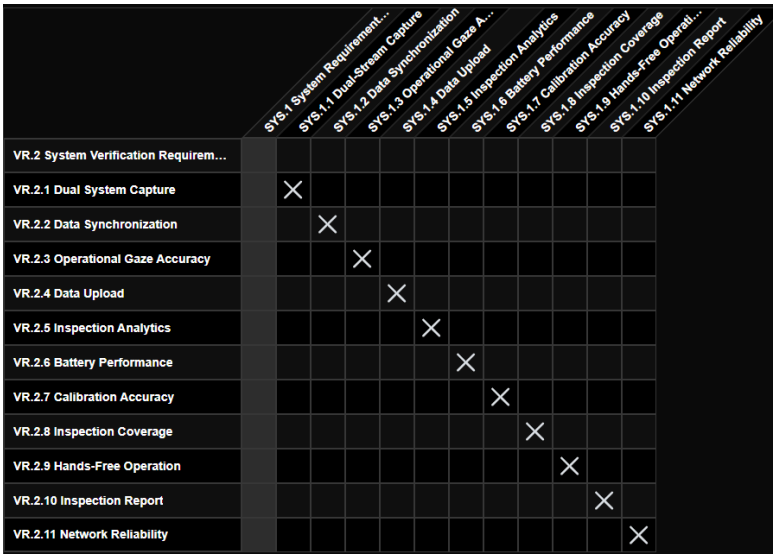
Rationale:

Quality Score: 0%

Labels: *None to display.*

Traceability matrix:

Traceability Matrix.csv



Attaching the picture of the traceability matrix from Innoslate since the table wasn't entirely visible in the final document.

	SYS.1	SYS.1.1	SYS.1.2	SYS.1.3	SYS.1.4	SYS.1.5	SYS.1.6
VR.2							
VR.2.1		X					
VR.2.2			X				
VR.2.3				X			
VR.2.4					X		
VR.2.5						X	
VR.2.6							X
VR.2.7							
VR.2.8							
VR.2.9							
VR.2.10							
VR.2.11							